

TECHNICAL DATA SHEET (TDS) – PRODUCT 2

PRODUCT NAME: Maintenance-Free Anti-Roll Bar Repair Bushing (Undersize Bore)

APPLICATION: BMW E38 7-Series Front / Rear Stabilizer Bar (Sway Bar Wear Compensation)

OEM BASE REFERENCE: 31351091555 (Standard 20.50 mm Specification)

1. DIMENSIONS & GEOMETRY

- **Outer Diameter (OD) & Width:** Strictly matching OEM factory tolerances to ensure a 100% tight fit inside the original BMW steel mounting brackets without any axial or radial housing play.
- **Standard Nominal Bore (For New/Unworn Bars):** 20.50 mm ID (Inner Diameter).
- **Repair / Undersize Bore Options (For Worn/Pitted Bars):** 20.30 mm / 20.10 mm / 19.90 mm ID
- **Bore Sizing Logic:** The internal bore diameter is reduced in **0.20 mm steps** to clamp tightly onto worn or rust-pitted stabilizer bars, while the external dimensions remain identical to the OEM standard.
- **Design:** Slotted split-axial configuration for quick installation.

2. MATERIAL SPECIFICATIONS

- **Body Compound:** High-Grade EPDM (Ethylene Propylene Diene Monomer) according to ASTM D2000.
- **Hardness:** 72 ±3 Shore A (Optimized to maintain original BMW luxury NVH vibration dampening and ride comfort; completely eliminates polyurethane harshness).
- **Self-Lubricating Liner:** Co-vulkanized PTFE (Polytetrafluoroethylene / Teflon) Shaved Liner/Mesh.
- **PTFE Layer Thickness:** 0.35 mm (Molecularly bonded to the EPDM internal bore surface during the high-pressure curing process).

3. PERFORMANCE & MAINTENANCE

- **Zero-Maintenance Operation:** 100% dry-fit system. **No external grease, silicone, or lubricant required** during installation or over the entire service life. Complete anti-squeak and anti-binding design.
- **Ozone Aging Resistance:** Must pass **ASTM D1149** (No micro-cracking after 72h exposure to 50 pphm ozone concentration at 80°C).



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